

Arith Mittel

$$\mu = \frac{x_1 + x_2 + \dots + x_n}{N} = \frac{\overset{\text{Anzahl 1en}}{n}}{N} = \Theta = \text{Fehlerrate}$$

$$\sigma^2 = \frac{(x_1 - \mu)^2 + (x_2 - \mu)^2 + \dots + (x_n - \mu)^2}{N} = \cancel{\times}$$

N ... Einträge gesamt davon n mit Wert 1

$N - n$ Wert 0

$$\textcircled{\times} \quad \frac{n(1-\theta)^2 + (N-n)(-\theta)^2}{N} =$$

$$\frac{n}{N} (1-\theta)^2 + \left(1 - \frac{n}{N}\right) \theta^2$$

$$= \Theta (1-\Theta)^2 + (1-\Theta) (\Theta)^2$$

$$= \Theta (1-\Theta) \underbrace{[(1-\Theta) + \Theta]}_1$$

$$\Rightarrow \sigma = \sqrt{\Theta(1-\Theta)}$$

$$\Rightarrow \frac{d}{d\theta} \Theta - \Theta^2 \stackrel{!}{=} 0$$

$$1 - 2\theta = 0$$

$$\theta \Rightarrow \frac{1}{2}$$

$$\sigma^2(\theta) \rightarrow \frac{d\sigma^2}{d\theta} \stackrel{!}{=} 0$$